Computer-aided Wartegg Drawing Completion Test

Lili Liu[1⋆][0000−0001−9558−4616], Graziella Pettinati[2][0000−0002−9405−4899] and Ching Y. Suen[3][0000−0003−1209−7631]

1. Centre for Pattern Recognition and Machine Intelligence (CENPARMI)
Concordia University, Montreal, QC, Canada
li_lil@encs.concordia.ca
2. Expertise Graziella Pettinati Inc. info@graziellapettinati.com
3.Centre for Pattern Recognition and Machine Intelligence (CENPARMI)
Concordia University, Montreal, QC, Canada
suen@cse.concordia.ca

Abstract. Wartegg Drawing Completion Test (WDCT) is one of the most commonly used personality analysis tools of graphology, which is the mapping from the inside world to personal qualities. It helps institutes or individuals to have a better knowledge of intrinsic personality characters. However, the WDCT evaluation of the applicants was manually performed by human experts, thus the accessibility and outcome of WDCT are heavily restricted by the availability and experience of the expert. To overcome such issues, this paper proposes the computer-aided WDCT (CA-WDCT) system, a fully-automatic WDCT system based on Digital Image Processing (DIP) and Machine Learning techniques. The CA-WDCT system extracts multimodal features and analyzes them under the Big-Five traits automatically. This CA-WDCT system can mitigate the heavy manual labour of psychologists and provide clients with flexible access.

Keywords: Personality Analysis · Wartegg Drawing Completion Test · Computer-aided Analysis

1 Introduction

The Wartegg drawing completion test (WDCT) was first proposed by the Germany psychologist Ehrig Wartegg in 1939 [1]. It has been developed and utilized widely in German-speaking countries and Latin America. However, it is not so popular in English-speaking countries [2]. In particular, in North America the first WTDC manual was published by Kinget in 1952 [3]. Based on the work of Kinget, Italy researcher Crisi extended and validated the Wartegg test in1998 and 2016 subsequently [4], [5]. The form of WDCT consists eight boxes as shown in Fig 1.

⋆ Corresponding Author

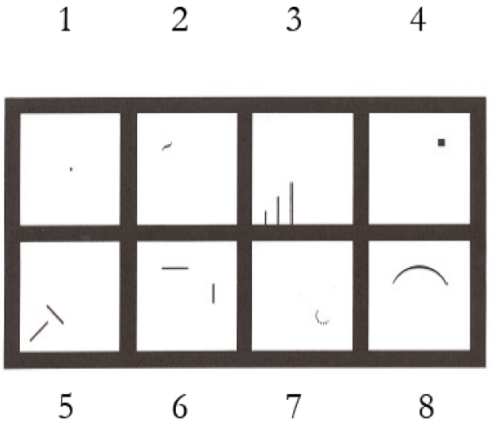

Fig. 1. The WDCT test form.

The WDCT shows its ability in a great range of application fields. It has been used in sociability evaluation by Gardziella in 1985 [6]. Gregory and Pereira have tried to imply WDCT in the personnel selection during recruitment process [7], [8]. In 2003, Deinlein used WDCT for further vocational counselling [9]. It can be also used in the field of education [10], [11], couple and family relationship therapy and inside condition diagnosis of patients [12], [13]. The WDCT shows its ability in a great range of application fields.

However, the form evaluation during the application of Wartegg test was done by psychologists by hand rather than automatically or electronically. Only three works have tried to exploit the computer aided analysis system implication. In 2008, Campos [14], the first try of organizing wartegg test through a software, was created to analyse test results and also to provide a person's personality. It transformed the WDCT form into an electronic version, which reduces the process time by 60%. Fast Wartegg Analyzer Tool (FWAT) is another kind of web-based application designed to preparing the origin image data for further exploration faster and easier, which is created by Yuanna in 2015. The speed of evaluation process accelerated by 65% of times compared to manual operation [15]. In 2018, Yuana promoted an approach based on the FWAT to calculate the cosine-similarity between the ideal characters and candidates' characters in order to select a perfect employee [16]. Though there are some attempts in the computer-aided applications for WDCT evaluation, they had only explored to display the form digitally and generate electronic results with extracted features instead of fully automatically evaluation. In order to achieve the target of improving the evaluation process of WDCT, we propose the computer-aided WDCT (CA-WDCT), a fully-automatic WDCT framework based on Digital Image Processing (DIP) and Machine Learning techniques. This CA-WDCT system can free the labour of psychologists and make it possible for any person in the world to access the WDCT at any time.

2 Methodology

The CA-WDCT analysis process is described in four steps as shown in Fig 2. First of all, we need to isolate the eight single boxes of the wartegg test form. Then, five predefined features are extracted automatically from each box by com-

puter program. Once the features are extracted from drawings, character analysis can be processed for a better understanding of hand-drawings and corresponding personalities based on the Big Five personality traits. Finally, the CA-WDCT system generates personality evaluation within a professional psychology report.

Image Isolation The first step of this project is to isolate the Wartegg test form from the scanned questionnaire and then segment it into eight boxes. We proposed a scissor algorithm for extracting the eight single boxes from questionnaires. The algorithm extracts the external contours with a modified template matching method to find out rectangle contour, which is hardly changed with the picture's affine transformation and its rotated angle. The distortion of the rectangle can be corrected with the angles. Within the cropped form, the system calculates the pixel distribution for each column and row to find out the watershed of border and drawings. Then, the "scissor" starts to work, cutting the form aligning the watershed like a paper cutting scissor into eight single boxes and stored following their numerical order. This method keeps a 100% of cutting accuracy even with rotated and messy strokes without clear boundaries.

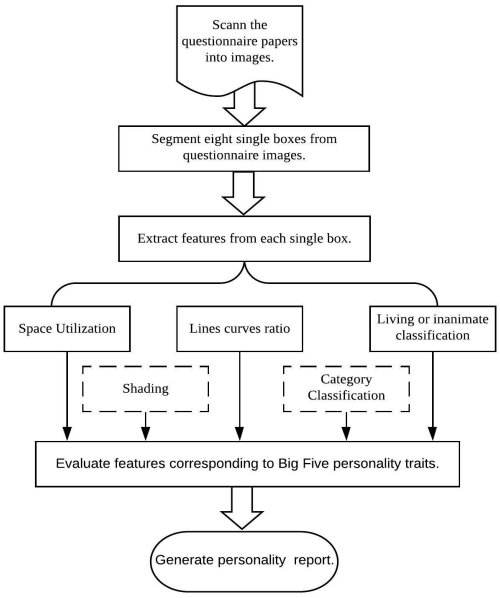

Fig. 2. The flowchart of this paper.

Feature Extraction Feature extraction is the most important part of CA-WDCT system. It makes fully-automatic analysis possible without human intervention. Moreover, the extracted features form the key clues for personality evaluation and report generation. Once we get the single box of drawing, the data for further steps are prepared well. As shown in Fig 2, we have five features needed to be extracted from each box. They are space utilization, line curves ratio, living or inanimate drawings classification. The first three features are implemented in this paper, and two further exploit features shading and category classification.

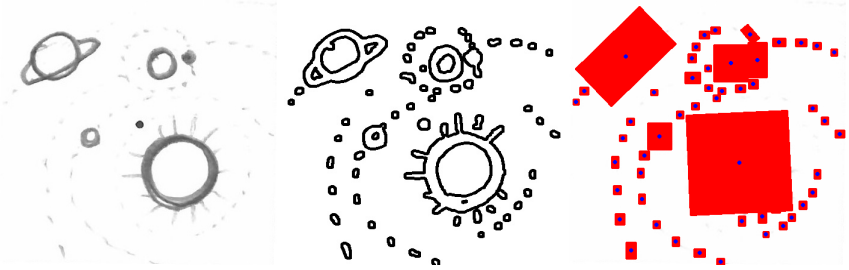

Fig. 3. Left: original picture. Middle: contoured picture. Right: utilized space.

Fig. 4. Left: original picture.Middle: colored line and curves.Right: connected line and curves.

Space Utilization and Distribution The first feature we have exploited is space utilization and drawings distribution. As shown in Fig 3, the software extracts the contour and center of each drawing, aligns the center covering each abject drawing with a red rectangle. The space utilization can be obtained by calculating the ratio between the red rectangle area and the whole boxes area. The distribution is calculated with the distance of each object drawing. The distance can be calculated from the location of each center point.

Lines and Curves Ratio The line-curve ratio feature is a kind of typical personality clue of drawings. Unlike human understanding, when two lines connect with each other, it is easier for the program to regard them as one object, which is actually should be treated as two. The system was programmed to find out the cross corner of two or more lines and split all lines apart. Then trying to connect the crossed lines back with extra heuristic information or principal component analysis. This two-stage cut-and-connect operation splits the connected lines separately with understandable meanings.

Living and Animate Objects In this part, we extracted the drawing features of their living or inanimate category. A special convolutional neural network Lenet-5 [17] is used for classifying living and animate drawing objects. The first

step is to augment the data with four different augmentations: Image Sharpening, Unsharp Mask, Adaptive Threshing and Binary. The ratio of training: validation: testing is 6 : 2 : 2. Images are resized into 28×28 and normalization is applied. Lenet-5, a classical model of deep neural networks, has been commonly used in deep learning area especially in handwriting classification. The model is a 6 layer convolutional (Conv) neural network (CNN) with repeated the architecture of Conv, Relu and Pooling for twice and following a fully-connected layer and softmax function at the end. The model details are shown in Table 1. In order to improve the result, we used the Adam optimizer, Mini-batch and early stopping method while training the model. With this CNN model, we got the training accuracy of 98.46% and testing accuracy of 86.15%.

3 Discussion & Future Work

For now, we have discussed the first several steps in this CA-WDCT system research. This work has finished isolating the eight boxes from the questionnaire images and extracted three features (space utilization, living and animation classification, lines-curves ratio). In the future step, at least two more features will be exploited and explored. Once the features are extracted from drawings, character analysis can be processed for a better understanding of hand-drawings and corresponding personalities based on the Big Five personality traits. The Big The big five personality traits are Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism. The CA-WDCT will analyze the above characteristics of each box and score them. The scoring results and the distribution of the eight boxes' scores are then mapped to the big five personalities analysis. The CA-WDCT system can generate a psychology report. This CA-WDCT feature extraction and analysis method can mitigate the heavy manual labour of psychologists and benefit clients with easier access.

Table 1. Lenet-5 model used in this paper.

Layer	Filter Shape	Activation function
Conv2D	(32, (7, 7))	relu
MaxPooling2D	(2, 2)	-
Conv2D	(64, (5, 5))	relu
MaxPooling2D	(2, 2)	-
Conv2D	(64, (3, 3))	relu
Flatten+Dense	(64)	relu
Dense	(2)	sigmoid

4 Acknowledgement

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